



Graphs in Machine Learning

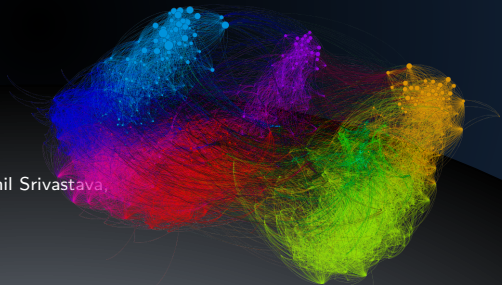
Distributed Graph Processing

Challenges and Requirements

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Partially based on material by: Rob Fergus, Nikhil Srivastava,
Yiannis Koutis, Joshua Batson, Daniel Spielman



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Large graphs do not fit in memory

Get more memory

Many challenges

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- ↳ minimize pass over data / communication costs

Needs to be consistent

- ↳ updates should propagate properly

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Different choices have different impacts: for example splitting the graph according to nodes or according to edges.

Many computation models (academic and commercial) each with its pros and cons

- MapReduce

- MPI

- Pregel

- Graphlab**

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<https://misovalko.github.io/mva-ml-graphs.html>